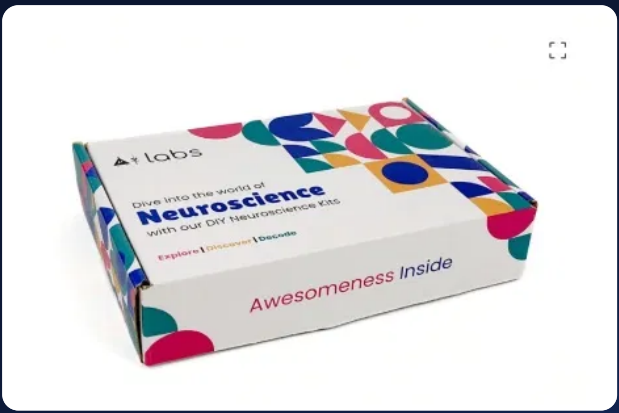
AVISHKAAR MAKEATHON 2026 · HEALTH GUARD

NeuroRide

Step-by-step Build Plan

From zero to a working eye-controlled smart wheelchair





Bhuvan Gowda BC
Bengaluru · Build window: 19 Oct – 30 Nov 2026

OVERVIEW

The plan at a glance

Buy in stages, prove each stage works, then move on. Never spend on the next stage until the current one passes its test.

Stage 0	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Now – 18 Oct	19 – 25 Oct	~26 – 31 Oct	Early Nov	Mid Nov	Before 30 Nov
Free prep: Wokwi, chassis, PPT story	Signals: blink, glance, clench	Motors + app joystick	Signal drives the chair	Health + emergency + flood	Video, PPT, submit

> **Deadlines:** 30 Oct bonus is a stretch goal. 30 Nov (Stage 1 submission) is the real target. Finale in Delhi: 15–16 Jan 2027 if selected.

Shopping list for Stage 1 (current cart)

Item	Status
DIY Neuroscience Kit – Basic (with UNO R4, ₹6,499 — UNO is a backup)	In cart
ESP32-WROOM-32 38-pin dev board	In cart
Female–Female jumper wires (you already have more wires)	In cart
Soldering iron 25W + flux	In cart
Solder wire — **swap 35% tin for 60/40**	Fix before checkout
USB data cable for ESP32 (data, not charge-only)	Check at home



What's inside the DIY Neuroscience Kit – Basic

STAGE 0 · NOW – 18 OCT · ₹0

Prep and build the chassis

Goal: a wheelchair-shaped body and an ESP32 you already know how to program.

A. Learn the ESP32 for free

1. Open **wokwi.com** → new ESP32 project.
2. Make an LED blink on GPIO 2.
3. Read a potentiometer on GPIO 34 and print it to Serial Plotter — this is exactly how we'll read the brain/eye signal later.
4. Install Arduino IDE + ESP32 board package at home.



Your Avishkaar Pro kit chassis — the starting body

B. Turn the Avishkaar chassis into a wheelchair

1. **Base frame** — rectangle of long beams, about 15 × 20 cm. Keep it flat and strong.
2. **Rear drive wheels** — 2 motors with the biggest wheels (differential drive).
3. **Front** — 1–2 small caster wheels or a ball caster.
4. **Seat** — flat plate raised 3–4 cm above the base.
5. **Backrest** — vertical beams + plate, leaning slightly back.
6. **Armrests + footrest** — short beams on the sides, small plate at the front bottom.
7. **Electronics bay** — space under the seat for ESP32, driver and batteries.

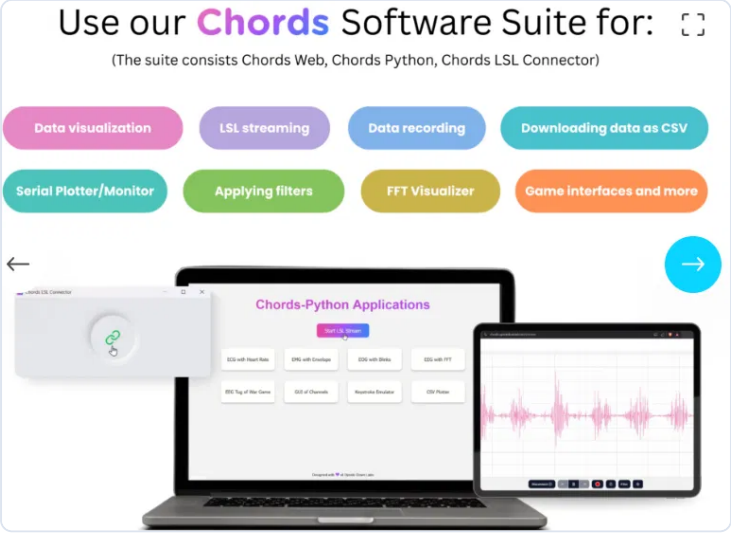
Sensor	Where to mount
HC-SR04 ultrasonic	Front, facing forward
MAX30102 heart / SpO2	Armrest, under the user's fingertip
MPU6050 tilt	Centre of the base, flat
Water sensor	Bottom, close to the ground
Buzzer + LED	Back of the backrest, visible

> Keep weight low and centred so it doesn't tip on turns. **Photograph every step** — you need them for the "build journey" slides.

STAGE 1 · 19 – 25 OCT

Read the eye and muscle signals

Goal: blink, glance and jaw clench detected correctly at least 8 times out of 10.



Day 1: check your signals in Chords before writing any code

Steps

1. **Day 1 — unbox and test with Chords:** plug the BioAmp EXG Pill into the UNO R4, open chords.upsidedownlabs.tech, and blink. If you see clear spikes, the kit works. If not, claim the 365-day warranty right away.
2. **Electrode placement (EOG):** one electrode beside each eye (left and right temple) for glance; one above an eye for blink; reference behind the ear. Clean skin with NuPrep first.
3. **Move to the ESP32:** wire the pill as below and flash the Upside Down Labs ESP32 EOG filter sketch.
4. **Write the detector:** double blink = start/stop, glance left/right = turn, jaw clench = stop.
5. **Calibrate:** 30-second routine that records each person's blink strength and sets thresholds.
6. **Test:** 10 tries of each gesture. Pass = 8/10 or better.

Wiring: BioAmp EXG Pill → ESP32

Pill pin	ESP32 pin	Note
VCC	3V3	Use 3.3 V, never 5 V, on the ESP32
GND	GND	
OUT	GPIO 34	ADC1 pin — keeps working when Bluetooth/WiFi is on

> Run the ESP32 from a laptop or power bank while testing signals — **not** from the motor battery. Keep electrode wires short and twisted.

> ✔ **Stage 1 passed when:** each gesture prints the right command on Serial Monitor 8+ times out of 10.

STAGE 2 · ~26 – 31 OCT · ~₹150

Drive the motors

Goal: the chair drives forward, back, left, right and stops from a phone.

1. Send me a photo of the Avishkaar motor connector and the Maker Board 2.0.
2. **Option A (default):** ESP32 → TB6612 (or L298N) motor driver → Avishkaar motors.
3. **Option B:** ESP32 → Maker Board 2.0 → motors, only if it accepts external commands.
4. Write motor functions: forward, back, left, right, stop, with speed control.
5. Add a simple Bluetooth joystick from the phone to test driving.

Wiring: TB6612 motor driver → ESP32

Driver pin	ESP32 pin	Purpose
PWMA / PWMB	GPIO 25 / GPIO 26	Left / right motor speed
AIN1, AIN2	GPIO 27, GPIO 14	Left motor direction
BIN1, BIN2	GPIO 16, GPIO 17	Right motor direction
STBY	GPIO 4	Enable driver
VM	Motor battery +	Separate motor supply
VCC, GND	3V3, GND	Common ground with ESP32

Two batteries: one for motors, one for ESP32 + BioAmp. Connect only their grounds together.

✔ **Stage 2 passed when:** the phone joystick drives the chair smoothly in all directions and stops instantly.

STAGE 3 · Early Nov — Signal drives the chair

1. Combine Stage 1 and Stage 2 code: gestures now move the chair.
2. Test with motors running and fix any noise (filtering, wire routing, battery separation).
3. Add the ultrasonic safety stop: obstacle under 20 cm = forced stop, overrides every command.

✔ **Stage 3 passed when:** you can drive a short course using only your eyes, and it always stops before a wall.

STAGE 4 · MID NOV · ~₹600–900 + perfboard

Health Guard, emergency and flood mode

Goal: the chair monitors health and raises the alarm by itself.

Buy now

MAX30102, MPU6050, 2× HC-SR04, water sensor, buzzer, LED, small perfboard (for 3.3 V and GND rails), optional SIM800L.

Wiring: sensors → ESP32

Part	ESP32 pin(s)	Note
MAX30102 + MPU6050	SDA GPIO 21, SCL GPIO 22	Both share the I2C bus
HC-SR04 front	TRIG GPIO 5, ECHO GPIO 18	ECHO is 5 V → use a voltage divider
Water sensor	GPIO 35	ADC1 input
Buzzer / LED	GPIO 23 / GPIO 19	
All 3.3 V and GND	Perfboard rails	One wire each back to ESP32

Features to build, in order

1. Heart rate + SpO2 readings with safe-limit alerts.
2. Fall/tilt detection → stop motors + start emergency check.
3. "Are you OK? Blink twice" check — no answer in 15 s = emergency.
4. Emergency alert chain: buzzer + light, SMS/call to family with location (simulated in the demo).
5. Auto doctor report: last 30 min of vitals + Medical ID.
6. Flood mode: water detected → lock movement + alert.

> Never dial 108 during testing. Simulate every emergency call in the demo.

> ✔ **Stage 4 passed when:** tipping the chair triggers the full alert chain, and dipping the water sensor locks the wheels.

STAGE 5 · BEFORE 30 NOV

Demo, pitch and submit

Goal: a 2–3 minute video and a PPT that make the judges remember NeuroRide.

1. **Prototype app UI** — Home, Control, Health Guard, Brain signal, SOS, AI assistant, Flood alert screens.
2. **Demo video** — eye-controlled drive → obstacle stop → fall → alert chain → flood lock.
3. **PPT** — problem, your personal reason, solution, how it works, what's new vs existing projects, build journey photos, results, future scope.
4. **Rehearse** the pitch until it fits the time limit.
5. **Submit** on the My Avishkaar portal with buffer days to spare.

Final checklist

- ☐ Eye/blink control works 8+ times out of 10
- ☐ Obstacle stop overrides every command
- ☐ Heart rate, SpO2 and fall detection working
- ☐ Emergency alert chain demonstrated (simulated)
- ☐ Flood mode locks movement
- ☐ App UI screens recorded
- ☐ Build photos collected
- ☐ Video and PPT done
- ☐ Confirmed with Avishkaar that non-Avishkaar parts are allowed
- ☐ Submitted before 30 Nov

> **Backup plan:** if the eye control acts up on stage, switch to app control. Health Guard, emergency response and flood mode still work — you never show a dead project.